JAYA PANGIHUTAN SITUMORANG

Full-Stack Engineer · AI Engineering

Medan, Indonesia · 0852-9756-0273 · jayasitumorang19@gmail.com · jayasitumorang.vercel.app · github.com/jayasitumorang

SUMMARY

Full-Stack Engineer with 5+ years of professional experience building web, mobile and enterprise systems for fintech, logistics, EdTech, HR and enterprise teams. Works across ASP.NET, Java Spring Boot, Node.js, Python, React, Next.js and React Native, on SQL Server, PostgreSQL and MySQL, with AWS, Azure and GCP. Builds production AI: live computer vision, data agents and self-hosted LLM tools (YOLO, Copilot Studio, vLLM, Ollama, RAG, OCR). Brings a security mindset from penetration testing and VAPT remediation.

WORK EXPERIENCE

Full-Stack Engineer · PT. Wilmar Consultancy Services Promoted

Apr 2023 – Present · Medan

TechDev AI

- Built live **computer vision** systems: worker safety-harness detection on an unloading line (YOLO + pose, ~7 fps), a weighbridge monitor with six automated weighing checks over multi-camera ZeroMQ streams, and a fruit bunch counter fed by Azure Blob Storage.
- Built an **enterprise data agent** in Microsoft Copilot Studio (GPT-4.1) for analysts and managers, answering only from internal SQL Server data, with strict no-assumption rules and Indonesian/English replies.
- Built self-hosted LLM tools on vLLM and Ollama: a private RAG document assistant, an OCR + LLM document reader, and a YOLO/SAM vision training studio.

Stack: Python, YOLO, OpenCV, ZeroMQ, Copilot Studio, vLLM, Ollama, FastAPI, SQL Server, Docker, Azure Pipelines

TechDev

- Maintain and extend enterprise systems from user requirements: Terminal Warehouse Management (TWMS) and Packaging Management (PMS) on Java Spring and RabbitMQ; HR System, e-Recruitment and Face Absence Management on ASP.NET.
- Upgraded legacy systems to newer stacks, resolved VAPT security findings, and delivered desktop and IoT apps.
- Closed Jira tickets ahead of target; set architecture and coding standards with the team.

Stack: .NET, Java Spring, RabbitMQ, SQL Server, Next.js, Kotlin, AWS, Azure DevOps

Full-Stack Engineer · PT. Saba Semesta Perkasa (Tunas Logistik)

Aug 2022 – Apr 2023 · Jakarta

- Built a web and mobile logistics platform from scratch: real-time tracking, invoicing, master data and role-based access control.
- Defined the architecture and coding standards, managed team tasks and estimates, and met users to solve field problems.
- Delivered a scalable, secure and well-documented system that the next team can maintain.

Stack: Next.js, React Native, GraphQL, PostgreSQL, Figma

Full-Stack Engineer · PT. Mega Harapan Mulia

Jan 2022 – May 2022 · Jakarta

- Developed key features for the Kelas.com, Kelas.work and Kelas.Center learning platforms with UI/UX and project teams.
- Refactored legacy code into modular, maintainable parts and met weekly development targets.

Stack: Laravel, MySQL, AWS, Figma, Trello

Full-Stack Engineer (Intern) · PT. Finansial Integrasi Teknologi (Pinjam Modal)

Oct 2021 – Dec 2021 · Tangerang

- Built a B2B loan and assurance transaction dashboard with clear data summaries, replacing manual paperwork with paperless flows.

Stack: React, Node.js, TypeScript, PostgreSQL

Full-Stack Engineer · Desa Karang Bangun

Jul 2020 – Oct 2020

- Delivered the village website end to end with the local government, from hosting setup and UI design to the platform.

Stack: WordPress, MySQL, HTML, CSS

SKILLS

Languages & frameworks: TypeScript, JavaScript, Java, C#, Python, Kotlin, PHP · React, Next.js, Vue.js, Angular, React Native, Spring Boot, ASP.NET, Node.js, Laravel, FastAPI

AI: YOLO, OpenCV, pose estimation, Copilot Studio, GPT-4.1, vLLM, Ollama, RAG, embeddings, OCR (Tesseract), SAM, LoRA fine-tuning, prompt engineering

Data & messaging: SQL Server, PostgreSQL, MySQL, NoSQL, GraphQL, REST API, RabbitMQ

Cloud & DevOps: AWS, Azure, GCP, Docker, Azure DevOps Pipelines, Git, Linux, Nginx

Security & QA: Penetration testing (Kali Linux, Nmap), VAPT remediation, Katalon Studio automation

Tools: Jira, Trello, Figma · **Languages:** Indonesian (native), English, Batak

EDUCATION

Telkom University, Bandung

2018 – 2022

Bachelor of Information Technology · GPA 3.37

Thesis: The Use of IoT in the Ad Satiation Method to Improve the Effectiveness of Common Carp Cultivation

SPEAKING & ORGANIZATIONS

Guest Speaker, AWS User Group Medan Meetup (2025)

Creative Division: Henceforth, PMK TES, L.O.H (2019–2020)

CERTIFICATIONS

- AI Productivity and AI API Integration for Developers, Hacktiv8 (2025)
- Speaker Certificate, AWS User Group Medan (2025)
- Certified Secure Computer User (2021)
- Problem Solving · SQL (Intermediate) (2021)
- The Complete 2024 Web Development Bootcamp (2021)
- React – The Complete Guide (incl. Next.js, Redux) (2021)
- TypeScript, Angular and Full Stack Java courses